

INDEPENDENT BA PORTFOLIO CASE STUDY - FITNESS STUDIO ERP - MAY 2026

KAGE Fitness Studio ERP - Business Rules Summary

A clear summary of the business rules that guide ERP behaviour across membership, PT sessions, trainer allocation, billing, freelancer management, renewals, maintenance, and reporting.

Project Type

Independent BA Portfolio
Case Study

Domain

Fitness Studio / ERP /
Gym Operations

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Portfolio Boundary Note

This is a fictional BA portfolio artefact inspired by real-world gym operations. It is not a live client project, production ERP system, or final technical implementation. The purpose is to demonstrate Business Analyst thinking through clear business rule documentation.

DOCUMENT PURPOSE

Why This Document Exists

This artefact defines the business rules that guide ERP behaviour and help stakeholders understand how decisions, validations, and workflows should work.

It supports portfolio presentation by showing that the system logic has been thought through in a structured, practical, and business-friendly way.

BUSINESS RULES OVERVIEW

What Business Rules Do

Business rules define how the ERP should behave in specific situations, such as duplicate member checks, PT balance updates, or payment validation.

They turn business policies into clear decision points that the system and users can follow consistently.

Why they matter: they reduce ambiguity, align stakeholders, support UAT, and make the workflow easier to explain.

In a BA portfolio, business rules show that the solution is practical, controlled, and grounded in real operational needs.

BUSINESS RULE SET 1

Membership Management

Business reason: Keeps member records clean and supports access, lifecycle and renewal visibility.

RULE ID	BUSINESS RULE	TRIGGER / CONDITION	SYSTEM ACTION	PRIORITY
BR-MB-001	Duplicate member profile should not be created with the same phone or email.	New member registration or profile update.	Block save and prompt user to check existing profile.	Must Have
BR-MB-002	MemberType must be selected during registration.	Member registration screen opens.	Require selection before save.	Must Have
BR-MB-003	MemberType allowed values are Direct Member, Freelancer, and Freelancer Client.	User selects MemberType.	Only accept defined values.	Must Have
BR-MB-004	Member status can be Active, Expired, On Hold, or Inactive.	Status is updated or displayed.	Restrict status to approved values.	Must Have
BR-MB-005	A member must be linked to a valid MembershipPlan.	Member profile is created or modified.	Block save if plan is missing or invalid.	Must Have

BUSINESS RULE SET 2

PT Session & Trainer Management

Business reason: Protects PT package accuracy, trainer continuity, schedule control and commission eligibility.

RULE ID	BUSINESS RULE	TRIGGER / CONDITION	SYSTEM ACTION	PRIORITY
BR-PT-001	PT session must be linked to Member, Trainer, and PTPackage.	PT session is scheduled.	Block save if any link is missing.	Must Have
BR-PT-002	Trainer allocation should consider availability, current PT load, preferred time slot, client goal, and trainer specialisation.	Admin assigns trainer to client.	Show suitable trainer options and warnings.	Must Have
BR-PT-003	The same trainer should continue with the PT client wherever possible for one package cycle or one month.	New PT allocation is requested.	Prefer existing trainer unless unavailable.	Should Have
BR-PT-004	Trainer double-booking should be blocked unless admin marks it as group/shared PT.	Two sessions overlap for the same trainer.	Block conflict or require override.	Must Have
BR-PT-005	PT package balance reduces only after SessionStatus = Completed.	Session is completed.	Reduce remaining package balance by 1.	Must Have
BR-PT-006	Cancelled sessions before start time should not reduce package balance.	Session is cancelled before start time.	Keep package balance unchanged.	Must Have
BR-PT-007	CompletedSessions should never exceed TotalSessions.	Session completion is recorded.	Block save if completion exceeds total.	Must Have
BR-PT-008	Trainer commission should be calculated only for completed eligible sessions.	PT session is completed and eligible.	Calculate commission for eligible session only.	Must Have

RULE ID	BUSINESS RULE	TRIGGER / CONDITION	SYSTEM ACTION	PRIORITY
BR-PT-009	CommissionStatus should remain Estimated until Admin/Finance approves the payout cycle.	Session commission is generated.	Keep status as Estimated until approval.	Must Have
BR-PT-010	Trainer and client may agree timing, but the system must still record the final scheduled slot.	Session timing is finalised.	Save the confirmed time slot in the schedule.	Should Have

BUSINESS RULE SET 3

Finance & Billing

Business reason: Improves payment accuracy, invoice traceability and revenue classification.

RULE ID	BUSINESS RULE	TRIGGER / CONDITION	SYSTEM ACTION	PRIORITY
BR-FN-001	Payment amount must not be blank, zero, or negative.	Payment is entered.	Block invalid payment save.	Must Have
BR-FN-002	Payment mode must be selected: Cash, UPI, Card, or Bank Transfer.	Payment is recorded.	Restrict mode to approved values.	Must Have
BR-FN-003	PaymentStatus allowed values are Paid, Partial, Pending, Overdue, and Refunded.	Payment status is updated.	Accept only approved values.	Must Have
BR-FN-004	InvoiceNumber must be unique.	Invoice is generated.	Block duplicate invoice number.	Must Have
BR-FN-005	InvoiceSharedStatus should track Not Shared, Sent, or Downloaded.	Invoice is issued or accessed.	Update invoice sharing status.	Should Have
BR-FN-006	Partial payment should show outstanding balance.	Amount paid is less than due amount.	Display remaining balance.	Must Have
BR-FN-007	PaymentPurpose should classify revenue correctly.	Payment is posted.	Require purpose selection.	Must Have

BUSINESS RULE SET 4

Freelancer Management

Business reason: Keeps KAGE facility revenue separate from freelancer private coaching income and supports monthly access-fee billing.

RULE ID	BUSINESS RULE	TRIGGER / CONDITION	SYSTEM ACTION	PRIORITY
BR-FR-001	Freelancer must also exist as a Member with MemberType = Freelancer.	Freelancer profile is created.	Block if linked member record is missing or wrong type.	Must Have
BR-FR-002	Freelancer client must also exist as a Member with MemberType = Freelancer Client.	Freelancer client is added.	Block if member type is invalid.	Must Have
BR-FR-003	Each FreelancerClient must be linked to one Freelancer.	Freelancer client record is created.	Require a single freelancer link.	Must Have
BR-FR-004	Freelancer billing is monthly.	Month-end billing is generated.	Create one billing cycle per month.	Must Have
BR-FR-005	MonthlyPayable = OwnFee + (ActiveClientCount × PerClientAccessFee).	Monthly billing is calculated.	Calculate payable using the monthly formula.	Must Have
BR-FR-006	Private PT/coaching income collected by freelancer is outside KAGE billing scope.	Freelancer has private coaching earnings.	Do not include in KAGE revenue.	Must Have
BR-FR-007	Do not use session-based billing or sessions × rate logic for freelancer billing.	Billing is calculated.	Prevent session-rate billing model.	Must Have
BR-FR-008	If freelancer monthly payable is overdue beyond threshold, AccessFlagStatus becomes Gate Flag Active.	Payment remains overdue beyond threshold.	Flag account for admin review.	Must Have
BR-FR-009	Admin can review or override gate flag.	Gate flag is active.	Allow admin decision to clear or keep flag.	Should Have

BUSINESS RULE SET 5

Renewals & Retention

Business reason: Creates a clear front-desk follow-up process while keeping WhatsApp/SMS automation outside the MVP.

RULE ID	BUSINESS RULE	TRIGGER / CONDITION	SYSTEM ACTION	PRIORITY
BR-RN-001	Renewal Due flag appears 7 days before membership expiry.	Expiry date is approaching.	Show renewal alert.	Must Have
BR-RN-002	Expired members should be marked Expired after expiry date if renewal is not completed.	Expiry date passes without renewal.	Update member status to Expired.	Must Have
BR-RN-003	Overdue renewal queue should show members who have not renewed after grace period.	Grace period ends.	Move member to overdue queue.	Must Have
BR-RN-004	Renewal follow-up should be assigned to front desk/admin.	Renewal queue is created.	Assign responsibility to front desk/admin.	Must Have
BR-RN-005	WhatsApp/SMS automation is outside MVP scope.	Renewal reminder is needed.	Use manual call, note, or follow-up state.	Must Have

BUSINESS RULE SET 6

Equipment Maintenance

Business reason: Supports repair traceability, replacement review and maintenance cost visibility.

RULE ID	BUSINESS RULE	TRIGGER / CONDITION	SYSTEM ACTION	PRIORITY
BR-EQ-001	Each equipment item must have a CurrentStatus.	Equipment record is created or updated.	Require status selection.	Must Have
BR-EQ-002	CurrentStatus values are Operational, Under Repair, Out of Service, and Decommissioned.	Status is selected.	Restrict to approved values.	Must Have
BR-EQ-003	Every maintenance log must link to one Equipment record.	Repair entry is created.	Block save without equipment link.	Must Have
BR-EQ-004	RepairCost must be zero or greater.	Repair cost is entered.	Block negative cost values.	Must Have
BR-EQ-005	ResolutionStatus values are Open, Under Review, Resolved, Pending Vendor, and Escalated.	Maintenance status is updated.	Restrict to approved values.	Must Have
BR-EQ-006	Repeated repairs within a defined period can trigger ReplacementReviewFlag.	Repair history exceeds threshold.	Flag equipment for review.	Must Have
BR-EQ-007	System only flags replacement review; final replacement decision is made by owner/admin.	ReplacementReviewFlag is set to Yes.	Do not auto-replace equipment.	Must Have

BUSINESS RULE SET 7

Reports & Analytics

Business reason: Keeps dashboards based on summarized operational data while respecting role-based access.

RULE ID	BUSINESS RULE	TRIGGER / CONDITION	SYSTEM ACTION	PRIORITY
BR-RP-001	RevenueReport is a reporting view, not a core transaction table.	Report is generated.	Summarise data only.	Must Have
BR-RP-002	MembershipRevenue is summarised from Payment records.	Monthly revenue report is created.	Pull membership-related payment values.	Must Have
BR-RP-003	PTRevenue is summarised from Payment / PTPackage records.	PT revenue report is created.	Aggregate PT-related income.	Must Have
BR-RP-004	FreelancerRevenue is summarised from FreelancerBilling.	Monthly report is generated.	Include freelancer billing totals.	Must Have
BR-RP-005	MaintenanceCost is summarised from MaintenanceLog.	Report is generated.	Include maintenance expenses.	Must Have
BR-RP-006	$\text{NetRevenue} = \text{MembershipRevenue} + \text{PTRevenue} + \text{FreelancerRevenue} - \text{MaintenanceCost} - \text{other recorded expenses}.$	Revenue summary is calculated.	Use approved business formula.	Must Have
BR-RP-007	Empty report export filter should be blocked.	User tries to export without selecting filter.	Show validation message and stop export.	Should Have
BR-RP-008	Reports should respect role access.	User opens revenue dashboard.	Show only authorised data by role.	Must Have
BR-RP-009	Trainer users should see assigned sessions and approved/estimated commission status only.	Trainer opens report screen.	Limit report visibility.	Must Have

RULE ID	BUSINESS RULE	TRIGGER / CONDITION	SYSTEM ACTION	PRIORITY
BR-RP-010	Owner/admin can see revenue, renewals, PT usage, and maintenance KPIs.	Owner/admin opens dashboard.	Display management-level KPIs.	Must Have

CRITICAL RULES

Key Business Rules

Freelancer billing

Billing is access-fee based, not session-based.

PT balance

Package balance reduces only after completed sessions.

Trainer scheduling

Double-booking is blocked unless shared/group PT is approved.

Scope control

WhatsApp/SMS automation is not part of MVP.

Reporting view

RevenueReport is a reporting view, not a transaction table.

Approval control

Final commission payout requires Admin/Finance approval.

RULE TO UAT

UAT Relevance

BUSINESS RULE AREA	EXAMPLE UAT SCENARIO	WHY IT SHOULD BE TESTED
Duplicate member control	Try creating a member with an existing phone number.	Checks duplicate prevention and record quality.
PT session completion	Mark a scheduled PT session as Completed.	Confirms package balance reduction and trainer commission logic.
PT cancellation	Cancel a session before start time.	Verifies that balance is not reduced incorrectly.
Trainer double-booking	Book overlapping sessions for one trainer.	Ensures schedule conflict control works.
Freelancer monthly payable	Change active client count and recalculate payable.	Confirms monthly billing formula is applied correctly.
Gate flag	Leave a freelancer payment overdue beyond threshold.	Validates admin review trigger logic.
Blank payment amount	Save a payment without entering an amount.	Tests mandatory field validation.
Empty report export	Attempt export without selecting a period.	Prevents invalid or useless report output.
Renewal due queue	Move a member near expiry and check queue visibility.	Confirms renewal follow-up timing is working.
Equipment replacement review	Enter repeated repairs for the same equipment.	Checks whether review flag is triggered correctly.

SCOPE NOTES

Assumptions and Scope

MVP-level rules: Rules are defined for BA portfolio demonstration and focus on business behaviour.

Configurable thresholds: Gate flag days and replacement review period can be configured by Admin.

Out-of-scope integrations: WhatsApp/SMS, payment gateway, biometric attendance, and GST filing are outside MVP.

MVP boundary

This is not a final technical specification. The document focuses on stakeholder clarity, business rules, and portfolio explanation instead of vendor integration or technical implementation design.

Business Rule Addendum - Trainer Role Access

Added to keep the Business Rules Summary aligned with the Live ERP Demo trainer dashboard and role-based login.

RULE ID	BUSINESS RULE	TRIGGER / CONDITION	SYSTEM ACTION	PRIORITY
BR-PT-011	Trainer users can view only their assigned PT clients, today's sessions, completed/pending sessions, progress notes, and their own commission status.	Trainer logs in or opens trainer dashboard/mobile view.	Show trainer-owned data only. Hide full gym revenue, freelancer billing, all-trainer payout data, and admin-only reports.	Must Have
BR-RA-001	Role-based access should control which modules each user can see.	User selects Owner/Admin, Front Desk, or Trainer demo role.	Owner/Admin gets full access; Front Desk gets operational entry views; Trainer gets PT and own session/commission views only.	Must Have

BA NOTE

The trainer dashboard is treated as a role-based view of existing PTSession, Trainer, Member, PTPackage and CommissionStatus data. It is not a separate payroll system, full mobile app, or new ERP module. This keeps the MVP practical while showing stakeholder access thinking. Full business revenue, freelancer billing and all-trainer payouts remain restricted to Owner/Admin/Finance roles.